



2 Numbers are stored in a computer using binary floating-point representation with:

- 10 bits for the mantissa
- 6 bits for the exponent
- two's complement form for both the mantissa and the exponent.

(a) Write the normalised floating-point representation of the following binary number using this system.

0.00000011010111

**Mantissa**

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**Exponent**

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|--|--|--|--|--|--|
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[2]

(b) Calculate the normalised binary floating-point representation of  $-25.3125$  in this system. Show your working.

**Mantissa**

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**Exponent**

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Working .....

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[4]

